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Cross-national evidence from real-world buildings

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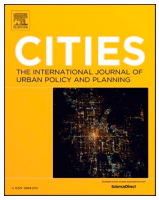
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Aesthetic preferences for traditional versus contemporary architecture: Cross-national evidence from real-world buildings

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ABSTRACT

Aesthetic experiences of the built environment represent an important pathway linking cities and subjective well-being. Despite growing interest in public evaluations of architectural design, existing research is largely based on single-country samples. This study aims to advance understanding of aesthetic preferences for building facades using survey data from two national contexts, Norway and Greece (N = 2997). It examines whether contemporary or traditional buildings are preferred and how sociodemographic and spatial characteristics relate to building preferences. Respondents in both countries evaluated the same four pairs of buildings, with each pair consisting of a contemporary and a traditional facade, both located in Norway. The results show that traditional architectural designs were preferred over contemporary architectural designs across all comparisons. This pattern was significant in seven out of eight pair tests. Public preferences for traditional architecture were similar across the two countries despite all buildings being located in Norway. Two of the four comparisons involved contemporary buildings that had received architectural awards or nominations, and traditional buildings were preferred over these professionally acclaimed buildings as well. In addition, women and residents of denser urban environments exhibited stronger preferences for traditional designs. Together, these findings raise important questions for architectural education, urban development practices, urban planning regulations, and public participation in shaping the qualities of the built environment.

1. Introduction

Affective experiences of the urban environment play an important role in human well-being (Mouratidis, 2018b, 2021, 2024). Building facades contribute to these experiences in both positive and negative ways by generating positive or negative emotional responses ranging from pleasant to unpleasant and from more to less stimulating (Brielmann et al., 2022; Nasar, 1994; Zeki, 2019). While emotional responses are an important dimension of how people experience architecture, empirical studies often rely on preference-based evaluations as a practical way to capture overall judgments of building facades. Existing research shows that architectural elements such as facade complexity rather than minimalism, symmetry rather than irregularity, and colorful rather than gray facades tend to be evaluated more favorably in terms of preference, aesthetic quality, and emotional response (Lindal & Hartig, 2013; Loodin & Thufvesson, 2022; Mouratidis & Hassan, 2020; Salingaros & Sussman, 2020; Stamps, 1999).

Despite growing interest in public evaluations of architectural design, empirical evidence remains fragmented. Most studies offer limited insight into how preferences vary across populations. Cross-national comparisons remain scarce as most studies come from a single national context (e.g. Herzog & Shier, 2000; Lindal & Hartig, 2013; Ro & Huffman, 2025; Stamps, 1999) and the influence of sociodemographic and spatial factors on building preferences has been explored only in a few studies (Loodin & Thufvesson, 2022).

This study provides knowledge that helps address these gaps by examining preferences for contemporary and traditional building facades using survey data from Norway and Greece (N = 2997). Respondents evaluated four pairs of real-world buildings, with each pair consisting of a contemporary and a traditional facade. The surveys presented photographs of eight buildings, all located in Oslo, the capital of Norway. By presenting respondents across two contrasting contexts, Norway and Greece, with the same set of buildings, the analysis enables a cross-national comparison of aesthetic preferences while holding the evaluated buildings constant. This approach allows the isolation of differences in preferences across populations. In this study, "contemporary architecture" refers to dominant design tendencies in current architectural practice, including minimalism, smooth surfaces, lack of

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ornamentation, asymmetrical and irregular shapes, and the use of industrial materials.

Drawing on this research design, the study makes three contributions. *First*, it adopts a cross-national perspective, by comparing how respondents from different national contexts evaluate the same architectural designs, thereby assessing the generalizability of preferences across populations. *Second*, it explicitly examines the role of sociodemographic and spatial characteristics, which have received limited attention in prior research. *Third*, the study includes award-winning and award-nominated contemporary buildings, enabling a preliminary empirical assessment of whether professionally acclaimed architecture aligns with people's perceptions. It thereby contributes to the literature on differences between professional and public evaluations of architecture.

2. Literature review

People tend to emotionally respond to the environment they are exposed to (Nasar, 1987, 1994; Posner et al., 2005; Russel & Pratt, 1980; Schindler et al., 2017). Emotional responses to the urban environment represent one of the main pathways between cities and subjective well-being (Mouratidis, 2018b, 2021, 2024). Emotional responses are lower in dense urban settings than in suburban or rural settings (Mouratidis & Yiannakou, 2022). Momentary happiness – how happy one feels at a particular moment – is lower in urban environments than in nature (MacKerron & Mourato, 2013). Urban environments of higher aesthetic quality could mitigate this handicap of the city versus nature (Negami et al., 2019; Seresinhe et al., 2019; Zeki, 2019). Urban aesthetic quality is not uniform across cities. It differs both within and across cities (Johansson et al., 2016; Salesses et al., 2013; Weber et al., 2008). For instance, some neighborhoods are perceived to be of higher aesthetic quality than others and this higher aesthetic quality is associated with higher neighborhood satisfaction (Mouratidis, 2018a).

Architectural design can influence emotional response to the urban environment by shaping building facades that in turn shape streetscapes and public spaces (Nasar, 1994). Momentary happiness was found to be higher when exposed to aesthetically pleasant streetscapes (Seresinhe et al., 2019). The design of facades influences emotional reactions during walking in cities (Brielmann et al., 2022; Brunner et al., 2025). Neurological research has shown that the experience of visual beauty increases activity in the emotional brain (part of the brain that regulates emotional and behavioral responses)– and in that way architectural design can influence individual health and societal well-being (Brielmann et al., 2022; Sussman & Hollander, 2021; Zeki, 2019). Therefore, some scholars have argued that beauty in architecture is not an added bonus but rather a necessity (Zeki, 2019) and that architectural styles should be shaped by public participation and should reflect residents' needs and preferences (Mouratidis & Hassan, 2020).

But what elements of buildings are visually appealing and how are contemporary trends in architecture perceived by the public? Research on perception and experience of architectural styles includes studies on preference, visual engagement, and emotional response. Preference and visual engagement studies reflect perceptions of architecture which may in turn trigger emotional responses (Brielmann et al., 2022; Nasar, 1992, 2008; Zhang & Lin, 2011).

Preference studies assess how different types of facades and elements of facades influence building preferences. An experiment that manipulated architectural design elements using computer-aided design found that adding architectural complexity to building facades (texture, ornament, cornice, door trim, and window trim) strongly increased preference (Stamps, 1999). A study of the facade of tall buildings found that building silhouette complexity was linked to higher perceived complexity, preference, and positive emotional response (Heath et al., 2000). Studies that compared traditional with modernist buildings found that traditional building facades were preferred when maintenance was accounted for (Herzog & Gale, 1996; Herzog & Shier, 2000). In a similar fashion, a study of preferences among city center managers found that traditional facades were perceived as more attractive than modernist ones (Loodin & Thufvesson, 2022). A study combining 360-degree videos with immersive virtual reality found that contemporary architectural styles were evaluated less positively than traditional styles (Mouratidis & Hassan, 2020). Moreover, traditional architectural styles may slightly increase acceptability for high-density urban development, according to a study in London (Navarrete-Hernandez et al., 2021). Finally, pastel colors have been found to be generally preferred to gray color palettes, typical of modernist and contemporary buildings (Strzałkowski & Lewicka, 2024).

Visual engagement studies attempt to predict eye fixation based on a scanned image. An engagement study of building facades with different window arrangements found that facades with more coherent and symmetrical window arrangements received higher scores in visual engagement than contemporary styles with asymmetrical and irregular arrangements (Salingaros & Sussman, 2020). Symmetry in architecture and urban design has been suggested as an important research area that requires further investigation (Mehaffy, 2020). Another study on visual engagement with building facades and public spaces found that eye fixation on traditional design is greater than with non-traditional design (Hollander et al., 2020). Finally, a recent visual engagement study assessed the results by National Civic Art Society (2020) that found approximately 72% of respondents in the United States preferring traditional designs for federal buildings over modern ones. The study carried out a series of visual attention experiments and confirmed that traditional buildings capture and retain respondents' attention to a greater extent than modern buildings (Ro & Huffman, 2025).

Emotional response studies investigate the impact of design on emotions and mood. A study on architectural variation and building height using computer-generated images found that architectural variation – ornamentation, trim, roof cornice – and lower building height improved emotional response and particularly restoration likelihood (Lindal & Hartig, 2013). In another study of emotional responses, traditional styles in architecture and public space design characterized by facade complexity, symmetry, and ornamentation were experienced as more relaxing than contemporary styles (Mouratidis & Hassan, 2020). This is in line with subsequent findings indicating that traditional architecture may result in increased perceived aesthetic quality (Sherborne & Malthouse, 2026) as well as reduced stress and mental fatigue for public space users (Brielmann et al., 2022).

Overall, existing evidence indicates that elements of traditional architecture including facade surface complexity, architectural variation, symmetry, and colorful palettes tend to be preferred, to be viewed as more attractive, and to generate more positive emotional responses in the built environment.

Another important body of research has examined *differences between professional and public perceptions of architecture*. Studies have found that architects and non-experts may systematically diverge in their aesthetic preferences, with professionals often favoring contemporary or avant-garde designs to a greater extent than the general public (Brown & Gifford, 2001; Hubbard, 1996; Šafárová et al., 2019). Several explanations have been proposed for this divergence, including differences in training, familiarity with architectural trends, and exposure to design norms that prioritize innovation, abstraction, and conceptual originality (Ghomeshi & Jusan, 2013; Gifford et al., 2002; McNeill, 2009).

Important knowledge gaps remain, nevertheless, within the topic of architectural preferences. Most existing studies rely on samples from a single country and there is a lack of cross-national evidence of public preferences. This limits insight into whether architectural preferences are context-specific or reflect broader patterns across different populations. Moreover, the influence of sociodemographic and spatial characteristics on architectural preferences is underexplored. Addressing these gaps is important to increase the robustness and generalizability of existing research. Finally, there is limited evidence on how professionally acclaimed contemporary architecture is perceived by the

public and whether it is aligned with public preferences. This is an important research avenue in the context of ongoing debates on contemporary architecture and urban development (Andersen et al., 2020; Andersen & Røe, 2017; Andersen & Skrede, 2025; Bern & Røe, 2022; Dreher et al., 2021; Mouratidis & Andersen, 2024; Ponzini, 2011; Tarazona Vento, 2017), divergence in architectural evaluations between architects and the public (Brown & Gifford, 2001; Šafárová et al., 2019), and discourses on contemporary versus traditional styles in architecture (Curl, 2018; McNeill, 2009; Spencer, 2016).

3. Data and methods

3.1. Data sources

This study uses data collected through questionnaire surveys in Norway and Greece. The two countries capture variation in architectural preferences across Northern and Southern European contexts, which differ in terms of urban form, architectural traditions, and exposure to contemporary architectural styles. This contrast provides a useful basis for examining whether aesthetic preferences for building facades are context-specific or generalizable across different populations. The survey data were collected as part of two broader research projects on the built environment and quality of life, with the present study focusing specifically on aesthetic evaluations of architectural design. Both questionnaires included identical questions on architectural preferences as well as individual sociodemographic attributes in order to be used for a comparative analysis. The size of the combined sample from the two surveys is 2997 respondents.

The Norwegian survey was carried out in Oslo and its surrounding region Viken. The survey was conducted during June–August 2020. The dataset is based on 1796 respondents aged 19–95. To draw the sample, the project team first obtained residential address data from the Norwegian Tax Administration. Using this list, 20,000 households were randomly selected and contacted by post with an invitation to complete an online survey. Participation was limited to one adult individual per household. Sampling was structured to reflect variation in both geography and urban form. The study area was divided into zones, and random selection was carried out within each. In total, 12,000 invitations were sent in Oslo (8000 in inner-city areas and 4000 in outer areas) and 8000 in Viken (equally split between suburban and more peripheral or rural locations). This design was intended to ensure representation across different densities and settlement types. Both the invitation and the questionnaire were available in Norwegian and English. The survey was pilot-tested and adjusted before its distribution, and no incentives were provided. After removing invalid addresses, the response rate was 9–10%, which suggests that non-response bias cannot be ruled out. The sample aligns with the broader population on sociodemographic factors such as age and gender, though some differences are observed, especially in terms of education level.

The Greek survey collected answers from the regions of Athens and Thessaloniki as well as various other regions across the country. The survey was conducted during April–May 2020. It was distributed through a campaign on social media – specifically 216 diverse local groups on Facebook, an article in an online magazine, and a project webpage – combined with snowball sampling. The sample size from this survey is 1201 respondents. The questionnaire was pilot-tested before distribution and was available only in Greek, which may have limited participation among non-Greek speakers. No incentives were offered. Although the sample reflects a broad mix of neighborhoods, urban forms, and socioeconomic conditions, it was not drawn from a formal sampling frame. Consequently, common survey biases are present, including overrepresentation of highly educated respondents and underrepresentation of people with migrant backgrounds. The implications of sample composition for interpretation are discussed in Section 5.

Measurement of all variables used in the study is presented in Table A1 in Appendix A. The dependent variable was measured as follows: “Please imagine a new building is planned to be built near where you live. Two different designs are proposed. Please look at the two buildings below. Which building would you most like to be built near you?” The question was asked four times, one for each of the four building pairs presented in the subsection below. This measure was used by a previous survey on preferences for building facades (YouGov, 2009). It captures respondents' overall evaluation of the buildings, as a proxy for preference and visual appraisal. Choosing a building to be located nearby implies a holistic evaluation that reflects, among other factors, the perceived aesthetic qualities of the facade as well as considerations associated with the surrounding environment. The results are interpreted as reflecting overall preference for building facades rather than purely affective responses.

Descriptive statistics of all explanatory variables are presented in Table 1, while descriptive statistics of building preferences are presented in the Results in Section 4. Both surveys have limitations including possible non-response bias and a high representation of highly educated individuals and residents of major cities compared to the population. To address this issue, statistical comparisons of architectural preferences are complemented with multivariable analyses that examine the influence of sociodemographic and spatial characteristics on architectural preferences.

3.2. Building pairs and architectural styles

To assess building preferences, the study examines four pairs of buildings presented in the survey as photographs (Fig. 1). Each pair consists of one building representing contemporary architecture and one building representing traditional architecture. The study focuses on building facades, so only the exterior of the building was shown in the photographs presented to respondents. All the buildings included in the study are from Oslo, Norway and the source of the photographs is Google Street View.

The sequence of building pairs in the questionnaire is that of Fig. 1, while the order in which photographs of contemporary and traditional buildings are presented within each pair corresponds to the order shown in Table A1 (Appendix A). A mixed (quasi-randomized) order of architectural styles was used, with the position of traditional and contemporary buildings varying across pairs, in order to reduce potential order and consistency effects. Survey questions, building pairs, photographs, and presentation sequences were identical in the Norwegian and Greek surveys.

Contemporary and traditional architecture encompass a range of contrasting stylistic elements. The contemporary buildings included in the study display several characteristics commonly associated with contemporary design, such as asymmetry, curves or irregular shapes, limited ornamentation, smooth facade surfaces, simplicity and minimalism, and the use of industrial materials such as glass and steel. Not all of these characteristics are present in every contemporary building examined. In contrast, traditional architecture is typically characterized by symmetry and regular forms, greater facade variation, ornamentation, and the use of softer materials and colors.

The eight buildings were selected from a broader set of buildings in Oslo visible through Google Street View. The selection was made by the research team following a purposive rather than random approach, and was guided by the aim of identifying building facades that represent recognizable characteristics of contemporary and traditional architectural styles. Comparability within each pair was ensured through careful visual matching of the selected facades. Buildings were chosen so that they were similar in terms of overall height category (mid-rise urban buildings), facade width as perceived in the frame, general massing, and viewing angle. All photographs were selected from Google Street View using comparable camera perspectives to minimize distortions related to distance and angle. This approach aimed to achieve perceived comparability between buildings within each pair, isolating stylistic differences as much as possible.

Table 1
Descriptive statistics of explanatory variables.

Variables	Norwegian survey				Greek survey			
	N	Min/Max	Mean	s.d.	N	Min/Max	Mean	s.d.
Age (years)	1796	19/95	49.67	16.50	1200	18/79	42.17	13.55
Female	1796	0/1	0.50	0.50	1201	0/1	0.56	0.50
Lives with partner or spouse	1796	0/1	0.69	0.46	1201	0/1	0.61	0.49
Higher education	1796	0/1	0.70	0.46	1201	0/1	0.71	0.46
Household with children	1796	0/1	0.33	0.47	1201	0/1	0.42	0.49
Religious	1796	0/1	0.07	0.26	1201	0/1	0.15	0.36
Distance to city center (km)	1796	0.40/217.17	17.56	27.13	999	0.16/56.60	8.96	7.80
Neighborhood density (people/ha within 1000 m radius)	1796	0.43/173.66	64.36	51.04	1159	1.00/247.00	95.09	62.98

The selection also aimed to include both relatively ordinary contemporary buildings and professionally acclaimed examples (i.e., award-winning or award-nominated designs). The inclusion of award-winning and award-nominated contemporary buildings is intended to examine whether highly regarded examples of contemporary architecture are aligned with public preferences, rather than to isolate the causal effect of professional recognition.

The selection process began with the identification of two professionally acclaimed contemporary buildings, chosen for their award-winning or award-nominated status (buildings A2 and A4, Fig. 1). Comparable (in height category, perceived width, general massing, and orientation) traditional buildings representing elaborate examples of traditional architecture were then identified using Google Street View in Frogner, an area of Oslo known for such buildings. This resulted in the selection of buildings B2 and B4. Next, two more ordinary contemporary buildings were identified using Google Street View in the area of Grünerløkka (buildings A1 and A3). Grünerløkka was then searched again using Google Street View to identify traditional buildings that were comparable to A1 and A3. This resulted in the selection of buildings B1 and B3. The number of candidate buildings considered at each step was not formally documented, which limits the extent to which selection effects can be assessed.

All the selected contemporary buildings were built in the 21st century, whereas the traditional buildings date from the late 19th century or early 20th century. Since their construction, no substantial alterations that would fundamentally change the architectural style of the facades were identified.

Overall, the methodological approach of the study does not produce a representative sample of buildings but enables a controlled comparison of contrasting architectural styles using real-world examples. Because the study relies on photographs of real-world buildings rather than digital designs, exact equivalence in physical dimensions cannot be guaranteed and some discrepancies between buildings are unavoidable.

Building pair 1: Contemporary building A1 was built circa 2015 and features largely asymmetrical volumes, lack of ornamentation, smooth surfaces, and a minimalist, industrial appearance. Conversely, traditional building B1 was built circa 1900 and has a symmetrical facade, with modest ornamentation and facade variation. It is located in Grünerløkka, a historically working-class area, and represents an example of relatively simple, less elaborate traditional architecture in Oslo.

Building pair 2: Contemporary building A2 is the Munch Brygge building, an example of high-end contemporary architecture in Oslo. It was completed in 2019 and is the winner of the Oslo Association of Architects Architecture Prize in 2019 and the Oslo City Architecture Prize in 2020. It has also received several positive reviews in some of Norway's largest magazines and newspapers. It is located in a highly prominent waterfront area of Oslo, adjacent to landmark buildings such as the Munch Museum, the Oslo Public Library, and the Oslo Opera House. It has an asymmetrical, minimalist, and smooth facade without ornamentation, with brick elements that create a warmer color palette than is typical for contemporary buildings. This building is compared with traditional building B2, located in Frogner, a historically affluent area and built in 1896–97. Traditional building B2 represents an elaborate example of traditional architecture in Oslo, characterized by a symmetrical facade with pronounced variation in texture and color, ornamentation, and overall complexity.

Building pair 3: Contemporary building A3, built in 2005, has a minimalist, smooth facade without ornamentation but is symmetrical in form. This is compared with a modest traditional building B3, built circa 1900, that has similar characteristics to B1: a symmetrical facade, with some ornamentation and facade variation. Both buildings A3 and B3 are found on the same street in Grünerløkka, a neighborhood that was historically working class but has undergone substantial gentrification.

Building pair 4: Contemporary building A4 is called the Munkedamsveien 62 building, built in 2015. It represents another example of high-end contemporary architecture in Oslo. It has been nominated for the ArcDaily's Building of the Year award in 2018, the European Union Prize for Contemporary Architecture – Mies van der Rohe Award in 2017, and the Oslo City Architecture Prize in 2016. It has a minimalist and smooth facade without ornamentation and a large proportion of glass. This building is compared with traditional building B4, built in 1898. Similar to B2, traditional building B4 is also found in Frogner, a historically affluent area, and represents an example of elaborate traditional architecture, with a symmetrical facade characterized by variation in texture and color, ornamentation, and overall complexity.

Although the study examines a limited number of buildings from a single city, the selected facades present recognizable characteristics commonly associated with contemporary and traditional architectural styles (e.g. minimalism, smooth surfaces, and irregular shapes versus ornamentation, facade variation, and symmetry). The selected buildings serve as illustrative examples that enable comparisons and are not intended as a representative sample of contemporary and traditional designs. The use of photographs of real-world buildings, rather than computer-generated images, enhances ecological validity by reflecting architectural forms that people encounter in everyday urban environments. Moreover, the analysis of preferences across two national contexts helps capture broader patterns of aesthetic evaluation across different populations.

3.3. Analytical methods

The analysis relies on the following statistical methods. First, a descriptive analysis of preference frequencies for each building pair was carried out. Frequencies are presented separately for the Norwegian and Greek survey to reveal possible context-based differences. To statistically assess whether differences in frequencies within pairs are statistically significant, binomial tests were carried out. The null hypothesis for the test assumed that contemporary and traditional buildings were equally preferred (proportion = 0.5).

The second part of the analysis models respondents' preferences for traditional versus contemporary buildings. Preferences were analyzed at the item level (building pair) using generalized estimating equations (GEE) with a binomial distribution and logit link. Each respondent contributed four observations corresponding to the four building pairs. Respondent identification number was specified as the clustering variable and building pair as the within-subject variable. An exchangeable



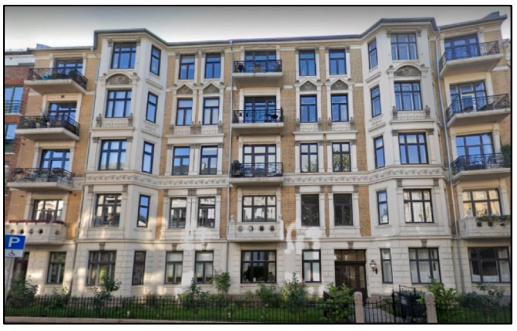
Contemporary building A1



Traditional building B1



Contemporary building A2



Traditional building B2



Contemporary building A3



Traditional building B3



Contemporary building A4



Traditional building B4

Fig. 1. Buildings compared in the study.

working correlation structure was used. Models included country, building pair, and sociodemographic and spatial characteristics as explanatory variables. Sociodemographic variables included age, gender, level of education, cohabitation status, the presence of children in the household, and religiosity. Income was not included in the models due to non-comparability in measurement scales between the Norwegian and Greek surveys. Migration-related variables were not included due to the number of migrant respondents in the Greek survey being too small for reliable estimation. Spatial characteristics were operationalized using continuous indicators – population density and distance to the

main city center – to capture urban–rural gradients in a comparable way across both countries. GEE modeling results are reported as coefficients and *p*-values. To assess whether the associations observed in the pooled GEE model were driven by specific building pairs, the analysis additionally includes separate binary logistic regression models for each pair. These results are used as robustness checks for the pooled GEE model, so the results are summarized in one table showing direction and significance only.

4. Results

4.1. Building comparisons

Binomial test results are presented in Table 2, while preferences and corresponding photographs of the building facades are presented in Fig. 2–5 to facilitate direct visual comparison. Since the study examines aesthetic evaluations, this format allows readers to relate reported preferences to the visual characteristics of the buildings being assessed.

For the first building pair, respondents in both countries preferred the traditional building (B1) over the contemporary building (A1) (Fig. 2). In Norway, 59% of respondents preferred B1 compared with 41% for A1, while in Greece the corresponding figures were 78% and 22%, respectively. The preference for the traditional building was statistically significant in both surveys (0.59, 95% CI [0.56, 0.61], $p < 0.001$ in the Norwegian survey and 0.78, 95% CI [0.75, 0.80], $p < 0.001$ in the Greek survey).

For the second building pair, respondents in both countries preferred the traditional building (B2) over the contemporary building (A2) (Fig. 3). In both Norway and Greece, 73% of respondents preferred B2 compared with 27% for A2. The preference for the traditional building was statistically significant in both surveys (0.73, 95% CI [0.70, 0.75], $p < 0.001$ in the Norwegian survey and 0.73, 95% CI [0.70, 0.75], $p < 0.001$ in the Greek survey).

For the third building pair, respondents showed a higher preference for the traditional building (B3) than for the contemporary building (A3) (Fig. 4). In Norway, preferences were nearly evenly split (52% versus 48%), whereas in Greece a clearer preference for the traditional building was observed (58% versus 42%). The preference for the traditional building was statistically significant only in the Greek survey (0.52, 95% CI [0.50, 0.54], $p = 0.114$ in the Norwegian survey and 0.58, 95% CI [0.55, 0.61], $p < 0.001$ in the Greek survey).

For the fourth building pair, respondents in both countries preferred the traditional building (B4) over the contemporary building (A1) (Fig. 5). In Norway, 82% of respondents preferred B4 compared with 18% for A4, while in Greece the corresponding figures were 83% and 17%, respectively. The preference for the traditional building was statistically significant in both surveys (0.82, 95% CI [0.80, 0.84], $p < 0.001$ in the Norwegian survey and 0.83, 95% CI [0.81, 0.85], $p < 0.001$ in the Greek survey).

When examining all four building pairs together, it is observed that preference for traditional architecture is very strong in building pairs 2 and 4 for both Norway and Greece. For building pair 1, preference for traditional architecture is very strong in Greece, while it is less strong but still significant in Norway. This is the pair where we observe the largest discrepancy between the two contexts, although preference still follows a similar direction in both surveys. For building pair 3, preference for traditional architecture is overall weaker than the other three pairs, but still significant in the Greek survey.

Table 2
Binomial tests and confidence intervals.

	Norwegian survey			Greek survey		
Probability (preference for traditional building)	Estimate	95% CI	p-value	Estimate	95% CI	p-value
Building pair 1 (B1)	0.59	[0.56, 0.61]	<0.001	0.78	[0.75, 0.80]	<0.001
Building pair 2 (B2)	0.73	[0.70, 0.75]	<0.001	0.73	[0.70, 0.75]	<0.001
Building pair 3 (B3)	0.52	[0.50, 0.54]	0.114	0.58	[0.55, 0.61]	<0.001
Building pair 4 (B4)	0.82	[0.80, 0.84]	<0.001	0.83	[0.81, 0.85]	<0.001

Notes: Two-sided binomial tests against a null proportion of 0.50. Estimates represent the proportion of respondents preferring the traditional building in each pair. Confidence intervals for proportions were calculated using exact (Clopper–Pearson) methods.

4.2. Modeling preference for traditional versus contemporary buildings

Table 3 presents the results from the GEE model predicting preference for traditional (versus contemporary) buildings. Preference for traditional buildings was coded as 1 and preference for contemporary buildings as 0. Respondent identification number was specified as the clustering variable and building pair as the within-subject variable. Positive and statistically significant coefficients indicate higher likelihood of preferring traditional buildings.

Table 3 shows that there is no overall significant difference in building preferences between respondents from Norway and Greece. It also indicates that preferences for traditional architecture vary across building pairs, consistent with higher proportions observed for some pairs in the binomial test results (Table 2). The GEE model indicates that the preference for traditional buildings is significantly stronger for building pair 4 than for pairs 1–3 (Table 3). In addition, both sociodemographic and spatial characteristics are associated with building preferences. Female gender and neighborhood density emerge as the strongest predictors, with women more likely than men to prefer traditional over contemporary buildings and higher neighborhood density associated with a greater preference for traditional architecture. Weaker but still statistically significant effects are observed for the presence of children in the household and higher educational attainment, both of which are associated with a higher likelihood of preferring traditional buildings.

Table 4 compares results from separate logistic regression models for each building pair. This serves as robustness check for the pooled GEE model of Table 3. Consistent with the GEE results, no overall systematic difference in building preferences between respondents from Norway and Greece is observed across pairs in Table 4. However, pair-specific models indicate that country differences emerge for building pairs 1 and 2, while no significant differences are found for pairs 3 and 4. Across all pairs, female gender and neighborhood density remain the strongest and most consistent predictors of preference for traditional buildings. The presence of children in the household is associated with a higher likelihood of preferring traditional buildings in pairs 1, 2, and 4, higher educational attainment is associated with preference for traditional buildings in pairs 3 and 4, while younger age is associated with preference for the traditional building in pair 3.

5. Discussion

5.1. Discussion of results

The study analyzed preferences for contemporary and traditional building facades using survey data from Norway and Greece. Overall, the findings provide clear evidence that traditional architectural designs are preferred over contemporary ones. Traditional buildings are preferred in all four pairs examined in the study. These preferences are statistically significant for all pairs in both the Norwegian and Greek surveys, with the exception of building pair 3, where the difference is significant only in the Greek sample. The observed preference for traditional over contemporary architecture is consistent with previous

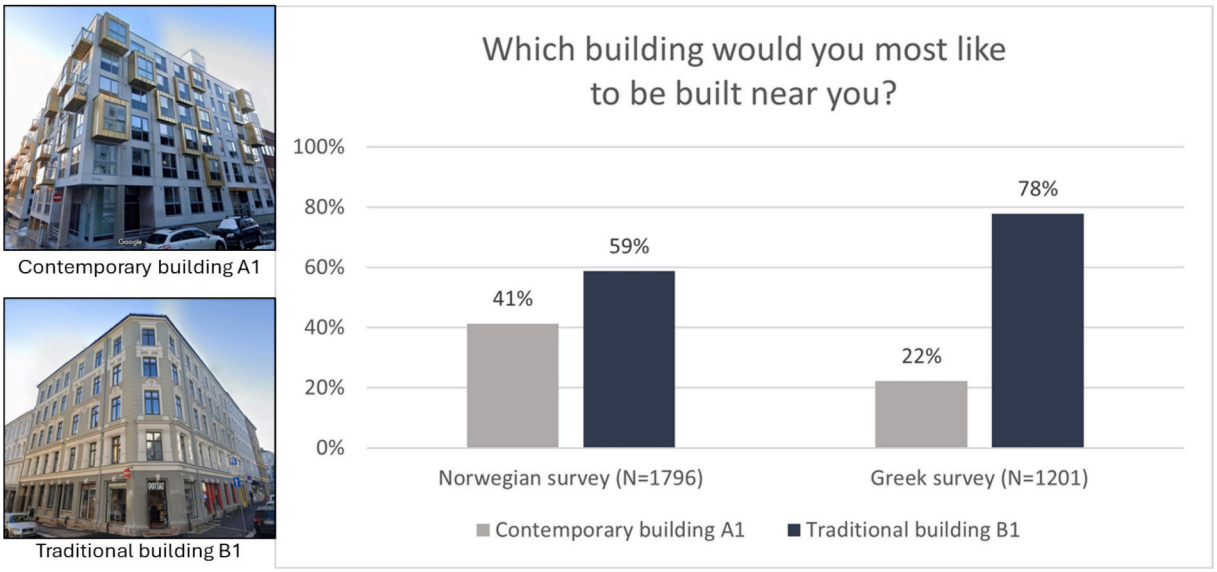


Fig. 2. Results from comparison between buildings A1 and B1. Binomial tests showed that preference for "Traditional building B1" was significant with $p < 0.001$ for both the Norwegian and the Greek survey.

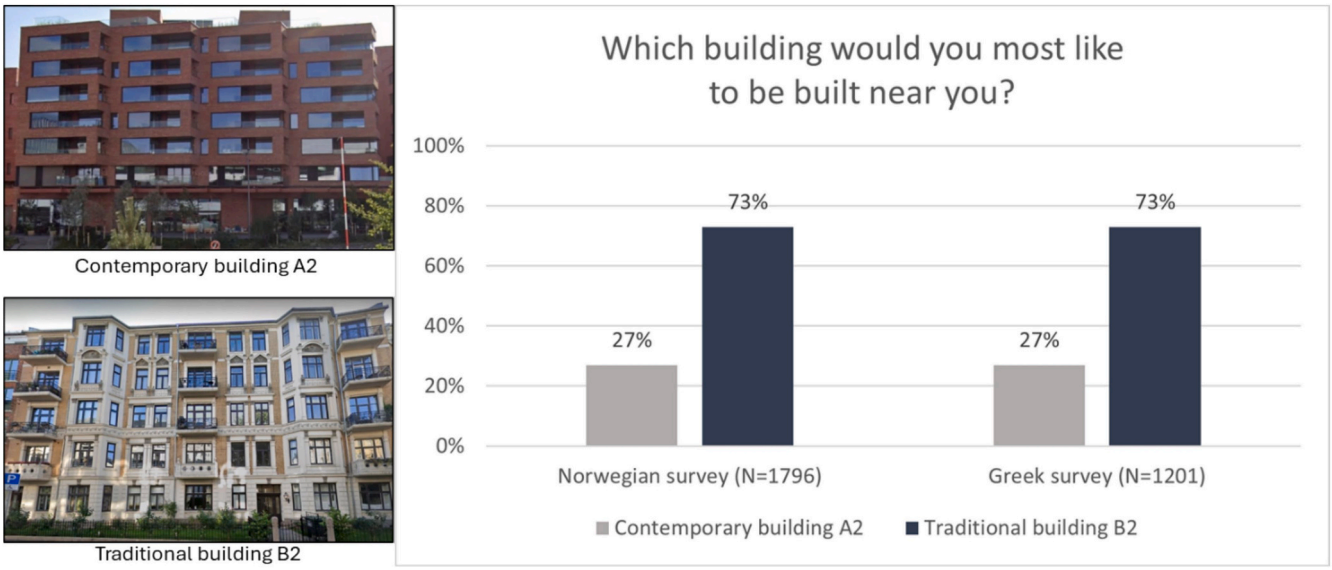


Fig. 3. Results from comparison between buildings A2 and B2. Binomial tests showed that preference for "Traditional building B2" was significant with $p < 0.001$ for both the Norwegian and the Greek survey.

research showing that traditional architecture is generally evaluated more positively (Herzog & Gale, 1996; Herzog & Shier, 2000; Loodin & Thufvesson, 2022; Mouratidis & Hassan, 2020). The findings are also broadly consistent with prior research showing that facade complexity, symmetry, ornamentation, and the use of color are often perceived as visually appealing and engaging (Lindal & Hartig, 2013; Salingaros & Sussman, 2020; Stamps, 1999; Strzałkowski & Lewicka, 2024). However, as these characteristics were not directly measured in the present study, their role should be interpreted with caution. More broadly, the results contribute additional empirical evidence to the literature on aesthetic experience in urban environments (Brunner et al., 2025; Johansson et al., 2016; Mouratidis, 2018b, 2021; Negami et al., 2019; Talen et al., 2023; Weber et al., 2008).

The finding that public preferences from Norway and Greece demonstrate similar patterns, despite buildings being situated in Norway, suggests that direct familiarity with specific buildings may play a more limited role than suggested in some previous studies (Cook & Furnham, 2012). This interpretation should be made with caution as some respondents from Norway may also be unfamiliar with the buildings. However, the similarity in responses across national contexts indicates that the observed preferences are not restricted to a single population. Norway and Greece differ substantially in their built environment, historical development, and architectural traditions. The consistency of results across these settings therefore suggests that the observed preference for traditional architectural forms reflects broader patterns rather than context-specific effects.

The results further show that traditional buildings are strongly preferred over the award-winning and award-nominated contemporary buildings (A2 and A4) examined in the study, with approximately 70–80% of the respondents in both Norway and Greece favoring traditional buildings. When comparing the two building pairs that include professionally acclaimed contemporary buildings with those including

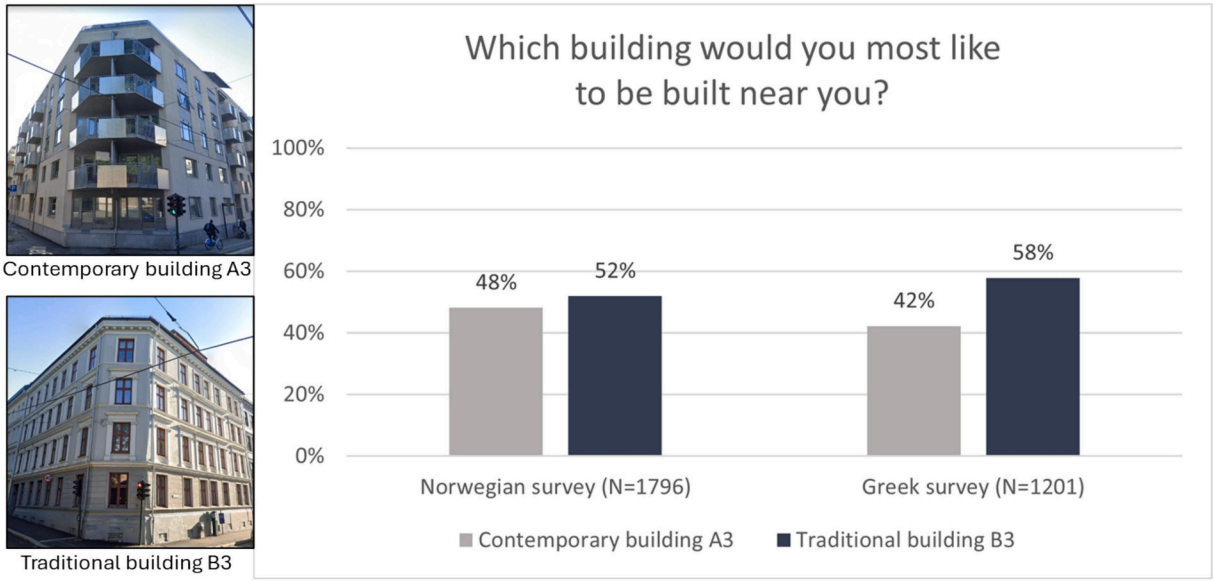


Fig. 4. Results from comparison between buildings A3 and B3. Binomial tests showed that preference for "Traditional building B3" was significant with $p < 0.001$ for the Greek survey but not significant with $p = 0.114$ for the Norwegian survey.

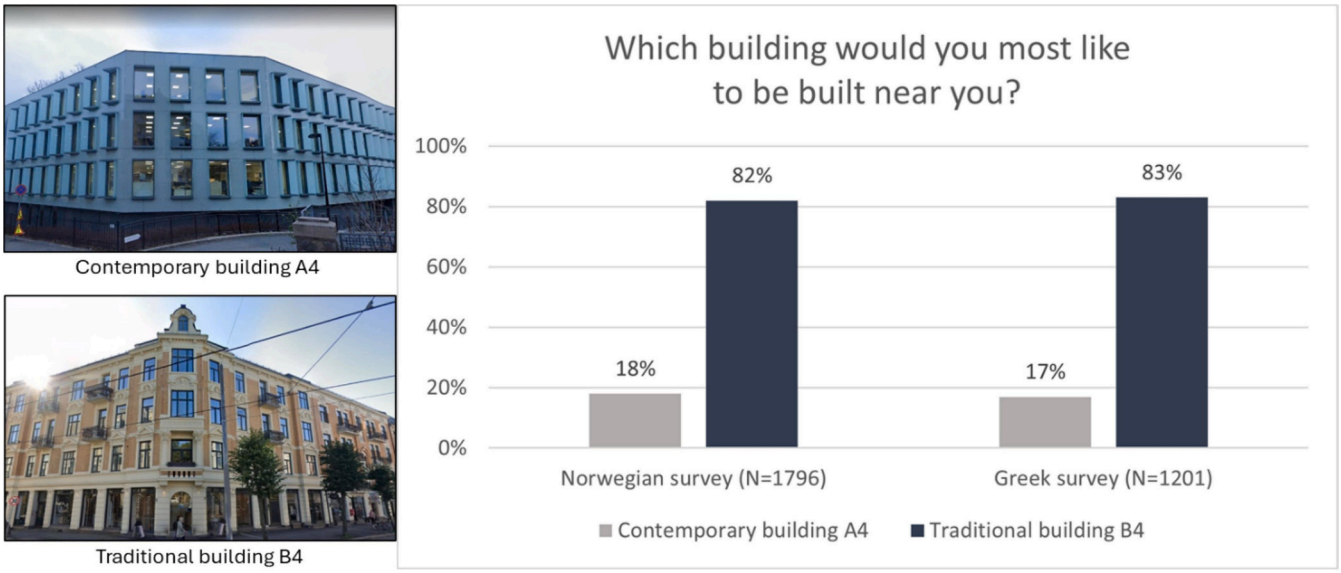


Fig. 5. Results from comparison between buildings A4 and B4. Binomial tests showed that preference for "Traditional building B4" was significant with $p < 0.001$ for both the Norwegian and the Greek survey.

more ordinary contemporary examples, preferences for traditional buildings are substantially stronger in the former case. This pattern may be influenced by differences in visual attributes of the traditional buildings included in the comparisons. Some buildings (e.g. B2 and B4) appear more elaborate in terms of ornamentation, color variation, and detailing, which may contribute to higher evaluations. However, these features were not systematically measured in the present study, and this interpretation should therefore be treated with caution.

The finding that traditional buildings are preferred over contemporary buildings that have received professional recognition suggests possible differences between public aesthetic preferences and prevailing professional evaluations within contemporary architecture. This interpretation should be made with caution, as the study does not examine the full set of buildings considered in award processes. Moreover, architectural awards may consider a broader set of criteria beyond facade aesthetics, including how buildings relate to their surroundings and their functional or conceptual qualities. We clarify that the present study focuses on visual evaluations of building facades and therefore does not assess these broader dimensions. The results, however, indicate that even professionally acclaimed contemporary buildings are not consistently preferred by the public when compared with traditional buildings.

The results raise the question of why contemporary architectural practice often differs from public aesthetic preferences. This question is relevant to discussions on how architectural aesthetics are assessed by professionals and experienced by everyday users. Several, possibly complementary, explanations can be considered. One explanation is that attributes such as facade complexity, ornamentation, colorful palettes, soft materials, and symmetry appear to be perceived as visually appealing by the public (Lindal & Hartig, 2013; Salingaros & Sussman, 2020; Stamps, 1999; Strzałkowski & Lewicka, 2024; Sussman & Hollander, 2021) but are often viewed as attributes of another era by

Table 3
Generalized estimating equation (GEE) model predicting preference for traditional (versus contemporary) buildings.

	Prefers traditional building (ref. contemporary building)		
Variable	Coefficient	p-value	Significance
Norwegian survey (ref. Greek survey)	−0.025	0.060	
Age	−0.001	0.113	
Female	0.068	0.000	***
Lives with partner or spouse	0.016	0.253	
Higher education	0.033	0.017	*
Household with children	0.040	0.003	**
Religious	−0.025	0.252	
Distance to city center	0.000	0.523	
Neighborhood density	0.001	0.000	***
Building pair 1	−0.168	0.000	***
Building pair 2	−0.098	0.000	***
Building pair 3	−0.283	0.000	***
Building pair 4 (ref.)	ref		
Intercept	0.718	0.000	***

Notes: $N = 11{,}164$. Number of subjects = 2791. Number of measurements per subject = 4.

Table 4
Coefficient comparison table: pair-specific binary logistic regression models predicting preference for traditional (versus contemporary) buildings.

	Prefers traditional building (ref. contemporary building)			
Variable	Building pair 1	Building pair 2	Building pair 3	Building pair 4
Norwegian survey (ref. Greek survey)	−***	+*	−	+
Age	−	−	−***	+
Female	+***	+***	+**	+***
Lives with partner or spouse	+	−	+	+
Higher education	−	+	+***	+*
Household with children	+*	+**	+	+*
Religious	−	−	−	−*
Distance to city center	+	+	+	+
Neighborhood density	+***	+***	+***	+***

Notes: $N = 2791$. Coefficient direction: + / −. Significance levels: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

architecture schools and the architect's profession in general and are often marginalized in development practices (Curl, 2018; McNeill, 2009). Another explanation concerns institutional and economic constraints, including development practices, cost and time considerations, and regulations, which may favor certain forms of contemporary design styles or simply may not encourage traditional designs. The present analysis cannot distinguish between these explanations, but it highlights the need for further research that explores the mechanisms that shape the differences between public preferences and built outcomes. Understanding these mechanisms is important for aligning architectural design practices more closely with human experience and well-being.

Considering the study's findings, the widespread dominance of contemporary architectural styles may raise concerns about their implications for everyday aesthetic experience and, potentially, for human well-being. In addition, the limited incorporation of public preferences into architectural decision-making processes highlights questions about democratic participation in the shaping of public spaces in the built environment. These reflections complement ongoing theoretical debates on contemporary architecture and renewed interest in traditional design principles (Curl, 2018; Mehaffy, 2020).

Finally, preferences for traditional buildings are found to be stronger among women and among respondents living in denser urban areas. The stronger preference observed among women suggests that perceptions of architectural styles and facade characteristics may differ by gender. The mechanisms underlying this association remain unclear. One possibility is that women respond differently to specific visual qualities of buildings, a hypothesis that requires further investigation. The finding that residents of denser urban environments show a stronger preference for traditional buildings is particularly relevant, as these individuals are more frequently exposed to multi-story buildings and dense urban fabrics in their everyday surroundings. By contrast, residents of suburban or rural areas typically encounter different housing typologies, such as detached or low-rise buildings, which may be experienced differently in terms of architectural style. Together, these findings are in accordance with previous research suggesting that gender and residential area may influence building preferences (Loodin & Thufvesson, 2022).

5.2. Limitations and future research

The paper has several limitations that point to avenues for future research. First, the analysis examines eight buildings grouped into four building pairs, which represents a small sample of facade examples. This means that the results may be influenced by the specific cases selected. The number of candidate buildings considered at each step was not formally documented, which limits the extent to which selection effects can be assessed. While the overall pattern is consistent across most comparisons and across both national samples, generalizing beyond the buildings included in the study should be done with caution. A greater number of buildings – including professionally acclaimed buildings – with varying architectural features could be examined in future studies.

Second, although efforts were made to ensure comparability within each building pair, some differences in height and overall size remain. This limitation could be addressed by including a greater number of building pairs or by employing digitally manipulated images or standardized architectural renderings.

Third, the study analyzes overall building preferences rather than discrete architectural elements such as symmetry, ornamentation, color, or texture. Such elements could be analyzed using computer-based manipulation in future studies.

Fourth, the study uses real-world photographs in a questionnaire format. This approach enabled the collection of a large number of responses in two countries. Future research could employ other methods such as virtual reality to provide a more immersive experience and an evaluation of emotional response to architectural styles.

Fifth, the use of real-world photographs introduces variation in contextual features such as lighting conditions, vegetation, and surrounding elements (e.g. parked cars), which may influence aesthetic evaluations. Such factors cannot be fully controlled in naturalistic stimuli. This represents a limitation of the study. However, the consistency of results across building pairs and across both national samples suggests that the observed preferences are not solely driven by these contextual differences. At the same time, the use of real-world images enhances ecological validity by reflecting how buildings are typically experienced in everyday contexts.

Sixth, the preference-based measure used in this study reflects overall evaluations of building facades and may incorporate considerations beyond immediate affective responses. Future studies could combine preference measures with measurements of affective responses.

Seventh, both surveys provide substantial sample sizes and cover diverse geographic areas but are not fully representative of the local populations. The Norwegian survey is subject to potential non-response bias due to the relatively low response rate, while the Greek survey relies on non-probability sampling, overrepresenting individuals who are more educated and digitally engaged. These characteristics are relevant for interpretation, as the analysis indicates that higher education is associated with a greater likelihood of preferring traditional architecture for some building comparisons (Table 4). The direction of potential bias is not straightforward and may reinforce the observed patterns to a certain extent. At the same time, the consistency of findings across two different sampling strategies and national contexts provides some

reassurance that the main results are not solely driven by sample composition.

6. Conclusions

This study has attempted to advance understanding of public preferences for contemporary and traditional architecture by using large survey samples from two national contexts and examining sociodemographic and spatial influences on building preferences. Overall, findings indicate a clear preference for traditional architectural designs over contemporary ones. In addition, award award-winning and award-nominated contemporary buildings received substantially lower preference ratings than traditional buildings in evaluations of preference, suggesting that architecture awards are not consistently aligned with public perceptions of aesthetic quality. Another key finding is that preferences for traditional architecture follow similar patterns in both national contexts examined, suggesting that familiarity with specific buildings or local building typologies does not appear to play a decisive role in building preferences. Finally, the study identifies female gender and residence in denser urban environments as consistent predictors of stronger preferences for traditional buildings.

These findings have important implications for research, practice, and policy. The observed gap between dominant architectural practices and public preferences raises important questions about how aesthetic considerations are addressed in contemporary urban development. This gap might be attributed to professional norms that prioritize innovation and conceptual expression as well as economic constraints, construction practices, and regulations favoring contemporary design trends or simply not encouraging the adoption of traditional design principles. From a research perspective, the results underscore the need for further empirical work on visual perception and aesthetic experience of architecture and urban design across different contexts and populations. From a practice and policy perspective, the findings call for reflection on current architectural and development paradigms. Architecture schools may benefit from reconsidering architectural styles or characteristics that are often viewed as outdated or obsolete in the architectural profession. Architects, developers, planners, and policymakers may benefit from engaging more closely with public preferences when shaping the visual qualities of the built environment, including building facades. Strengthening public participation and reconsidering regulations related to urban aesthetics may help foster built environments that are more closely aligned with users' experiences and preferences, thereby contributing to more inclusive and satisfying urban environments.

CRediT authorship contribution statement

Kostas Mouratidis: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cities.2026.107410.

Data availability

Data will be made available on request.

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